

BIOVOICEPRINT SCORING TREE

BASELINE CONTEXT LAYER

Pre-Recording Wellness Check-In

Question answered before each baseline audio session:

- balanced_centered
- mental_clarity_focus
- physical_energy
- recording_comfort_readiness
- restored_recovered

Each question is scored from 1 to 5.

Per-session outputs:

- wellness_anchor_total_score
 - Formula: sum of the five answers
 - Range: 5 to 25
- wellness_anchor_score_0_100
 - Formula: $(\text{wellness_anchor_total_score} / 25) * 100$
 - Direction: higher = stronger self-reported pre-recording wellness
- wellness_anchor_band
 - 85-100 = Strong Subjective Wellness
 - 70-84.99 = Generally Steady Subjective Wellness
 - 50-69.99 = Mixed Subjective Wellness
 - <50 = Lower Subjective Wellness

Baseline-level Wellness Anchor output:

- baseline_wellness_anchor_score_0_100
 - Formula: average of baseline sessions' wellness_anchor_score_0_100
- standard_deviation
 - Formula: standard deviation of baseline sessions' wellness_anchor_score_0_100
 - Purpose: supports future within-person standardized comparison
- score_range
 - Formula: lowest and highest baseline wellness-anchor score

- n_sessions
 - Number of baseline check-ins contributing to the baseline wellness anchor

Item-level baseline output:

Each of the five Wellness Anchor questions also stores:

- mean_response
 - Average 1-to-5 response across baseline sessions
- mean_normalized_score_0_100
 - Formula: $(\text{mean_response} / 5) * 100$
- standard_deviation
 - Standard deviation of that item across baseline sessions
- response_range
 - Lowest and highest 1-to-5 response for that item
- normalized_score_range_0_100
 - Lowest and highest item response converted to 0-100
- n_sessions
 - Number of baseline check-ins contributing to that item

Baseline BioVoicePrint output:

- baseline_biovoiceprint
 - This stores the person's baseline RDI distribution for future Wellness Congruence comparison.
- mean_rdi_score_0_100
 - Average baseline-session RDI score calculated against the final personal baseline reference
- rdi_standard_deviation
 - Standard deviation of baseline-session RDI scores
- rdi_score_range
 - Lowest and highest baseline-session RDI score
- n_sessions
 - Number of baseline sessions contributing
- session_rdi_scores
 - Session-by-session baseline RDI values

Score direction:

- Higher RDI = greater baseline-relative voice-pattern shift

Important boundary:

- The baseline wellness anchor does not score, change, or interpret the acoustic BioVoicePrint baseline. It stores the person's self-reported normal state beside the voice baseline for later comparison.
- The baseline_biovoiceprint block does not diagnose or interpret health status. It stores the person's normal RDI variation so future comparison sessions can compare current voice-pattern shift against the person's own BioVoicePrint baseline distribution.



STAGE 3

Raw Acoustic Feature Extraction

“What did the voice recording measure?”

Features extracted (19):

- f0_mean_hz
- hnr_db
- spectral_centroid_hz
- spectral_bandwidth_hz
- formant_f1_hz
- formant_f2_hz
- formant_f3_hz
 - Reference only features
 - Not used to for stage 4-7 scoring
- jitter_local
- shimmer_local
- zero_crossing_rate
- envelope_variance

- tremor_band_power_3_12hz
- approximate_entropy
- sample_entropy
- fractal_dimension
- lyapunov_proxy
- resp_sync_lag_sec
- resp_amp_variance
- phonation_stability_exhalation

Task-specific feature sources:

Sustained phonation only (4):

- jitter_local
- shimmer_local
- tremor_band_power_3_12hz
- phonation_stability_exhalation

Connected speech only (7):

- resp_sync_lag_sec
- resp_amp_variance
- envelope_variance
- approximate_entropy
- sample_entropy
- fractal_dimension
- lyapunov_proxy

All voiced tasks (5):

- f0_mean_hz
- hnr_db
- spectral_centroid_hz
- spectral_bandwidth_hz
- zero_crossing_rate

Reliability flags:

- valid = feature can be scored normally

- low_confidence = feature is kept, but should be interpreted carefully
- not_scored = feature is excluded from weighted averages

Reliability flags are currently applied to (6):

- jitter_local
 - Needs clear pitch cycles; zeros may mean poor detection, not true zero jitter
- resp_sync_lag_sec
 - Audio-only breath timing proxy; zero may mean no detectable lag, not perfect timing
- f0_mean_hz
 - Zero usually means pitch was not detected
- hnr_db
 - Depends on reliable pitch/periodicity; weak pitch makes HNR unreliable
- shimmer_local
 - Needs clean vocal amplitude cycles; zeros may mean detection failed
- phonation_stability_exhalation
 - Uses an estimated exhale proxy; weak envelope patterns reduce confidence

Note:

Stage 3 extracts raw measurements only. No severity score is assigned here.



STAGE 4A

Feature-Level Baseline Comparison

“How different is each feature from this person’s baseline?”

Each Stage 3 feature is compared to:

- baseline mean
- baseline standard deviation
- baseline min
- baseline max

Each feature receives (13):

- raw_delta
 - The direct difference between current value and baseline mean
- normalized_delta
 - The difference adjusted by baseline standard deviation
- z_score
 - Same as normalized delta; shows how many SDs from baseline mean
- abs_z_score
 - No sign attached to z-score
- sd_deviation_score_0_100
 - Z-score converted to a 0-100 shift score
- sd_severity_band
 - Severity label based on SD distance
- sd_color
 - Color based on SD severity
- range_deviation_score_0_100
 - Shift score based on personal min/max range
- range_band
 - Severity label based on min/max range shift
- range_color
 - Color based on min/max range shift
- final_deviation_score_0_100
 - Final feature shift score using the worse SD or range score
- Band
 - Final severity label
- Color
 - Final display color

SD/Z-score formula:

- $z_score = (\text{current value} - \text{baseline mean}) / \text{baseline standard deviation}$
- $abs_z_score = \text{absolute value of } z_score$
- $sd_deviation_score_0_100 = abs_z_score \times 25$

Final feature score:

deviation_score_0_100 = higher severity of:

- SD/Z-score shift
- personal min/max range shift

Important scoring rule:

- If a feature is marked not_scored, it is skipped from weighted averages. It is not treated as zero.

Severity scale:

- 0-12.5 = Baseline Consistent / green
- 12.5-25 = Minimal Shift / light green
- 25-37.5 = Mild Pattern Shift / yellow
- 37.5-50 = Moderate Pattern Shift / orange
- 50-75 = Significant Pattern Shift / red
- 75-100 = Extreme Pattern Shift / deep red



STAGE 4B

Domain Grouping (7)

“Which broader voice-pattern areas shifted?”

1. Variability Pattern Shift

Features:

- a. jitter_local: 20%
- b. shimmer_local: 20%
- c. envelope_variance: 20%
- d. approximate_entropy: 20%
- e. sample_entropy: 20%

2. Harmonic Structure Shift

Features:

- a. hnr_db: 35%
- b. spectral_centroid_hz: 20%

- c. spectral_bandwidth_hz: 20%
- d. f0_mean_hz: 25%

3. Transition Continuity Shift

Features:

- a. resp_sync_lag_sec: 40%
- b. resp_amp_variance: 25%
- c. phonation_stability_exhalation: 35%

4. Micro-Modulation Shift

Features:

- a. tremor_band_power_3_12hz: 65%
- b. lyapunov_proxy: 35%

5. Signal Clarity Shift

Features:

- a. hnr_db: 50%
- b. zero_crossing_rate: 25%
- c. spectral_bandwidth_hz: 25%

6. Distribution Balance Shift

Features:

- a. spectral_centroid_hz: 35%
- b. spectral_bandwidth_hz: 35%
- c. f0_mean_hz: 30%

7. Sustained Phonation Shift

Feature:

- a. phonation_stability_exhalation: 100%

Note:

If a feature is not_scored, it is skipped from that domain's weighted average.



STAGE 4C

Overall RDI Score (regulation deviation index)

“What is the total baseline-relative voice-pattern shift?”

RDI Score = weighted average of the 7 Stage 4 domains:

- Variability Pattern Shift: 20%
- Harmonic Structure Shift: 15%
- Transition Continuity Shift: 15%
- Micro-Modulation Shift: 15%
- Signal Clarity Shift: 10%
- Distribution Balance Shift: 10%
- Sustained Phonation Shift: 15%

RDI direction:

Higher = greater baseline-relative voice-pattern shift

RDI band:

- 0-12.5 = Baseline Consistent / green
- 12.5-25 = Minimal Shift / light green
- 25-37.5 = Mild Pattern Shift / yellow
- 37.5-50 = Moderate Pattern Shift / orange
- 50-75 = Significant Pattern Shift / red
- 75-100 = Extreme Pattern Shift / deep red



STAGE 5

Higher-Level Pattern Translation (5)

“What larger voice-pattern summaries can be built from Stage 4?”

1. Activation Pattern Variability

Inputs:

- a. Variability Pattern Shift: 40%
- b. Micro-Modulation Shift: 35%
- c. Signal Clarity Shift: 25%

2. Regulation Consistency Shift

Inputs:

- a. Harmonic Structure Shift: 35%
- b. Transition Continuity Shift: 35%
- c. Sustained Phonation Shift: 30%

3. Flexibility Range Shift

Inputs:

- a. approximate_entropy deviation score: 25%
- b. sample_entropy deviation score: 25%
- c. fractal_dimension deviation score: 25%
- d. lyapunov_proxy deviation score: 25%

4. Return-to-Baseline Window Shift

Inputs:

- a. RDI Score: 40%
- b. Transition Continuity Shift: 30%
- c. Sustained Phonation Shift: 30%

5. Adaptive Transition Pattern Shift

Inputs:

- a. Transition Continuity Shift: 35%
- b. resp_sync_lag_sec deviation score: 25%
- c. resp_amp_variance deviation score: 20%
- d. phonation_stability_exhalation deviation score: 20%

6. Overall Voice Pattern Shift

Inputs:

- a. Activation Pattern Variability: 20%
- b. Regulation Consistency Shift: 25%
- c. Flexibility Range Shift: 15%
- d. Return-to-Baseline Window Shift: 20%

e. Adaptive Transition Pattern Shift: 20%

Stage 5 direction:

Higher = greater baseline-relative voice-pattern shift



STAGE 6

Longitudinal Tracking

“How are comparison results changing over time?”

Inputs tracked over repeated comparison sessions:

- RDI Score
- Overall Voice Pattern Shift
- Regulation Consistency Shift
- Return-to-Baseline Window Shift
- Adaptive Transition Pattern Shift

Outputs:

- comparison_session_count
- rdi_trend_slope
- overall_pattern_shift_slope
- regulation_consistency_shift_slope
- return_to_baseline_window_shift_slope
- adaptive_transition_pattern_shift_slope
- stability_window_status
- variability_cluster
- trend_note

Note:

Stage 6 does not create new acoustic features. It tracks whether Stage 4 and Stage 5 scores are changing across comparison sessions.



STAGE 7A

Customer / Dashboard Output

“How should the result be summarized?”

Customer-facing scores are reversed so higher is easier to understand.

1. Regulation Stability

Inputs:

- a. RDI Score: 50%
- b. Regulation Consistency Shift: 30%
- c. Signal Clarity Shift: 20%

Display formula:

- Regulation Stability Score = $100 - \text{Regulation Stability Shift}$

2. Adaptive Capacity

Inputs:

- a. Variability Pattern Shift: 30%
- b. Flexibility Range Shift: 35%
- c. Adaptive Transition Pattern Shift: 35%

Display formula:

- Adaptive Capacity Score = $100 - \text{Adaptive Capacity Shift}$

3. Recovery Consistency

Inputs:

- a. Return-to-Baseline Window Shift: 45%
- b. Transition Continuity Shift: 30%
- c. Sustained Phonation Shift: 25%

Display formula:

- Recovery Consistency Score = $100 - \text{Recovery Consistency Shift}$

4. Voice Pattern Stability

Input:

- a. Overall Voice Pattern Shift: 100%

Display formula:

- Voice Pattern Stability Score = $100 - \text{Overall Voice Pattern Shift}$

5. Baseline Alignment

Input:

- a. RDI Score: 100%

Display formula:

- Baseline Alignment Score = $100 - \text{RDI Score}$

Customer-facing score direction:

- Higher = closer to personal baseline

Customer strength labels:

- 90-100 = Strong
- 80-89 = Stable
- 65-79 = Watch
- 50-64 = Variable
- Below 50 = Shifted



STAGE 7B

Wellness Congruence

Question answered: “How does the person's pre-recording self-report compare with the baseline-relative BioVoicePrint pattern?”

- This layer compares the five-question Wellness Anchor with the BioVoicePrint RDI pattern after each measure has first been compared to its own personal baseline. It does not change the BioVoicePrint score.

Important scoring correction:

- The Wellness Anchor score and BioVoicePrint Baseline Alignment score are not treated as equivalent just because both are displayed on a 0-100 scale.
- The primary comparison is now:
 - current Wellness Anchor vs baseline Wellness Anchor
 - current RDI vs baseline BioVoicePrint RDI distribution
- The raw display gap may still be shown descriptively, but it is not interpreted as a physiological discrepancy.

Current session wellness anchor:

- score_0_100
 - Current comparison session's wellness_anchor_score_0_100
- raw_total_0_25
 - Current comparison session's five-question total
- band
 - Current subjective wellness band
- items
 - The five current Wellness Anchor item scores stored separately
 - balanced_centered
 - mental_clarity_focus
 - physical_energy
 - recording_comfort_readiness
 - restored_recovered

Each item stores:

- raw_score_1_5
- normalized_score_0_100
- item_pattern
 - Descriptive comparison of each current item against that item's baseline mean
 - Purpose: identifies the lowest or most changed Wellness Anchor item without mapping it directly to a voice feature

Baseline wellness anchor:

- score_0_100
 - Average wellness-anchor score from baseline sessions
- standard_deviation
 - Standard deviation of baseline wellness-anchor scores
- band
 - Baseline subjective wellness band
- score_range
 - Baseline self-report low-to-high range
- n_sessions
 - Number of baseline self-report sessions contributing

Current BioVoicePrint:

- rdi_score_0_100
 - Current Regulation Deviation Index
- Direction
 - lower = closer to personal voice baseline
- rdi_band
 - Current RDI band
- baseline_alignment_score_0_100
 - Formula: $100 - \text{RDI}$
- Direction
 - higher = closer to personal voice baseline

Baseline BioVoicePrint:

- mean_rdi_score_0_100
 - Average baseline-session RDI score
- rdi_standard_deviation
 - Standard deviation of baseline-session RDI scores
- rdi_score_range
 - Lowest and highest baseline-session RDI score
- n_sessions
 - Number of baseline sessions contributing

Baseline-relative comparison:

- subjective_change_points
 - Formula: current wellness anchor - baseline wellness anchor
- Meaning
 - how much today's self-report differs from the person's usual baseline self-report
- objective_rdi_change_points
 - Formula: current RDI - baseline mean RDI
- Meaning
 - how much today's BioVoicePrint RDI differs from the person's usual baseline RDI pattern
- subjective_strain_z
 - Formula: (baseline wellness anchor mean - current wellness anchor) / baseline wellness anchor standard deviation
- Direction
 - positive = current wellness is lower than usual, meaning more self-reported strain
- Near zero
 - current wellness is about the same as usual
- Negative
 - current wellness is higher than usual, meaning the person reported feeling better than their own baseline
- objective_shift_z
 - Formula: (current RDI - baseline mean RDI) / baseline RDI standard deviation
- Direction
 - positive = greater voice-pattern shift than usual
- Near zero
 - usual personal BioVoicePrint state
- Negative
 - less shifted than usual BioVoicePrint state
- comparison_direction
 - Positive standardized values represent greater baseline-relative strain or voice-pattern shift.

Descriptive index comparison:

- index_gap
 - Formula: current wellness anchor - baseline alignment score

Important interpretation limit:

- The Wellness Anchor and BioVoicePrint alignment scores are different constructs. Their raw difference is descriptive only and is not treated as a physiological mismatch.

Alignment classifications:

- BROADLY_ALIGNED_STABLE
 - Label: Broadly Aligned and Stable
 - Meaning: The person feels close to their usual wellness baseline, and the voice pattern also remains within its usual baseline-relative variation.
- CONVERGENT_SHIFT
 - Label: Shared Baseline-Relative Shift
 - Meaning: Both the wellness check-in and BioVoicePrint changed in the same general direction. This is a pattern to monitor, not evidence of a cause or diagnosis.
- VOICE_PATTERN_LEADING
 - Label: Voice-Pattern Shift With Stable Self-Report
 - Meaning: The person feels close to their usual state, while the voice pattern shows greater-than-usual change. This may reflect early variation, recording context, vocal use, hydration, fatigue, or another unassigned influence.
- SELF_REPORT_LEADING
 - Label: Self-Reported Shift With Stable Voice Pattern
 - Meaning: The person reports feeling different from usual, while the voice pattern remains close to its normal range. The lived experience remains important even when it is not mirrored in this voice recording.
- DIVERGENT_DIRECTION
 - Label: Different Direction of Change
 - Meaning: The wellness check-in and BioVoicePrint moved in different directions. This difference should be observed over time rather than interpreted from one session.
- MIXED_PATTERN
 - Label: Mixed Baseline-Relative Pattern
 - Meaning: The two measurements show a mixed pattern that requires additional sessions for meaningful interpretation.
- DESCRIPTIVE_ONLY
 - Label: Descriptive Comparison Only

- Meaning: There are not yet enough stable baseline observations to standardize the comparison.

Comparison strength:

- Insufficient
 - Used when recording quality fails or a minimum personal baseline has not yet been established.
- Preliminary
 - Used when baseline exists but fewer than 5 paired comparison sessions are available.
- Developing
 - Used when 5-9 paired comparison sessions are available.
- Established Longitudinal Pattern
 - Used when 10 or more paired comparison sessions are available.

Important note:

- Comparison strength replaces unsupported numerical confidence percentages. It describes data maturity, not statistical proof.

Session quality:

- recording_quality_passed
- noise_floor_passed
- microphone_consistency_passed
- protocol_completion_passed
- vocal_discomfort_reported
- acute_context_flags
- interpretation_allowed

Available acute context flags:

- recent_strenuous_exercise
- high_vocal_use
- dehydration_possible
- poor_sleep
- caffeine_recent

- respiratory_symptoms
- different_recording_device
- unusual_environment

Important note:

- Context flags do not automatically change the score. They explain conditions that may affect interpretation.

Longitudinal alignment:

- paired_sessions_available
 - Number of paired Wellness Anchor and BioVoicePrint comparison sessions available
- trend_status
 - Whether enough paired sessions exist for trend interpretation
- subjective_trend
 - Current self-report trend status
- biovoiceprint_trend
 - Current BioVoicePrint trend status
- alignment_pattern
 - Current Wellness Congruence classification label
- voice_pattern_shift_with_stable_self_report_count
 - Number of sessions where the voice pattern showed greater-than-usual shift while self-report stayed close to baseline
- self_report_shift_with_stable_voice_pattern_count
 - Number of sessions where self-report shifted from baseline while the voice pattern stayed close to baseline
- minimum_sessions_for_trend
 - Minimum paired sessions needed before trend interpretation becomes meaningful
- trend_note
 - Do not interpret a single mismatch as physiology preceding awareness. That can only become a testable hypothesis after repeated paired sessions.

Output wording:

- summary

- Plain-language summary describing current self-report, baseline self-report, current RDI, baseline RDI, alignment type, and comparison strength.
- monitoring_note
 - Short reminder that interpretation becomes more meaningful when the same directional pattern repeats across valid paired sessions.
- boundary
 - One non-diagnostic boundary statement.



FINAL SIMPLE SUMMARY

Baseline Context Layer:

Collects five pre-recording self-report questions for each baseline session, creates the baseline wellness anchor, stores item-level wellness patterns, and stores the baseline BioVoicePrint RDI distribution for future comparison.

Stage 3:

Measures raw acoustic voice features.

Stage 4A:

Compares each feature to the person's personal baseline using Z-score and min/max range checks.

Stage 4B:

Groups feature shifts into 7 voice-pattern domains.

Stage 4C:

Combines the 7 domains into the RDI score.

Stage 5:

Translates Stage 4 domains into broader voice-pattern summaries.

Stage 6:

Tracks score changes across repeated comparison sessions.

Stage 7A:

Creates simplified customer/provider output with higher-is-better display scores.

Stage 7B:

Creates Wellness Congruence by comparing the current five-question Wellness Anchor to the person's baseline Wellness Anchor, and the current BioVoicePrint RDI to the person's baseline BioVoicePrint RDI distribution.

Boundary:

BioVoicePrint measures baseline-relative voice-pattern shift. Wellness Congruence compares self-report and voice-pattern change relative to each measure's own baseline; it does not diagnose stress, autonomic dysfunction, emotion, disease, or health status.